

Digital Asset Treasuries as Volatility Amplifiers: Empirical Evidence from Corporate Bitcoin Adoption

David Krause, Emeritus Associate Professor of Finance, Marquette University

Abstract

This study provides comprehensive empirical evidence that Digital Asset Treasury (DAT) companies, publicly traded firms holding significant cryptocurrency reserves, systematically amplify cryptocurrency volatility and create novel transmission channels between digital and traditional equity markets. Analyzing 250 trading days across seven DAT companies from November 2024 to November 2025, I document four key findings: (1) DAT stocks exhibit $1.95\times$ Bitcoin's volatility on average; (2) universal Bitcoin betas exceeding 1.0 across the DAT universe ($\beta = 1.34$ for Strategy); (3) $1.60\times$ greater price declines during crypto sell-offs; and (4) significant uncompensated idiosyncratic risk, comprising 52% of total DAT stock variance. Building on previous work that Bitcoin's Value at Risk (VaR) exceeds traditional treasury instruments by more than 125x, my findings reveal that DAT strategies compound these fundamental risks through structural amplification mechanisms. The results indicate that corporate cryptocurrency adoption establishes structural amplification mechanisms, magnifying market volatility, posing a risk to financial stability, and effectively exposing equity investors to hidden, leveraged cryptocurrency exposure. The findings have important implications for institutional investors, corporate treasurers, and financial regulators monitoring the integration of digital assets into traditional finance.

Keywords: digital asset treasuries, Bitcoin, volatility transmission, corporate finance, systemic risk, cryptocurrency

Digital Asset Treasuries as Volatility Amplifiers: Empirical Evidence from Corporate Bitcoin Adoption

1. Introduction

The emergence of Digital Asset Treasury (DAT) companies represents a fundamental shift in corporate finance, defined as publicly traded firms allocating a significant portion of their treasury reserves to cryptocurrencies, primarily Bitcoin. This strategy was pioneered by Strategy (formerly MicroStrategy), beginning with an initial \$250 million Bitcoin investment in August 2020. Since that time, the firm has consistently increased its holdings, accumulating a total reserve exceeding \$66 billion (Strategy, 2025). By 2025, this treasury model had been adopted by over 40 publicly listed entities, collectively holding approximately \$110 billion in Bitcoin (Bitcoin Treasuries, 2025).

Nevertheless, Krause (2025a) demonstrated through comprehensive Value at Risk analysis that cryptocurrencies introduce risk levels fundamentally incompatible with treasury management principles. He showed that Bitcoin exhibited daily Value at Risk (VaR) of -5.01% compared to just -0.04% for Treasury bills, a 125-fold difference. While framed as innovative treasury management and inflation hedging, DAT strategies create complex risk transmission channels between historically separate crypto and equity markets that remain empirically underexplored.

This paper provides the first comprehensive analysis of DAT companies as systematic volatility amplifiers. I address critical gaps in the literature by quantifying: (1) the magnitude of volatility amplification, (2) the universality of leveraged Bitcoin exposure across DAT types, (3) asymmetric behavior during market stress, and (4) the composition of DAT risk profiles. My findings emerge amid growing regulatory concern about digital asset integration in traditional finance (SEC, 2023; Basel Committee, 2022) and provide empirical grounding for ongoing policy debates.

The proliferation of DAT strategies represents a natural experiment in corporate risk-taking. Traditional corporate finance literature emphasizes conservative treasury management focused on liquidity preservation and risk minimization (Bates et al., 2009; Opler et al., 1999). DAT strategies fundamentally challenge these principles by allocating corporate reserves to one of the most volatile asset classes in modern finance (Urquhart, 2016; Dyhrberg, 2016). DAT companies transform treasury management from a stabilizing function into a major source of volatility, a shift that subjects institutional investors to hidden, leveraged crypto exposure due to structural risk amplification (Krause, 2025b). Understanding the implications of this shift is crucial for institutional investors, corporate boards, and financial regulators.

2. Literature Review

2.1 Corporate Treasury Management and Cash Holdings

Traditional corporate finance theory provides clear frameworks for treasury management. The trade-off theory of cash holdings suggests firms balance the marginal costs and benefits of liquid

assets (Opler et al., 1999), while the pecking order theory emphasizes financing hierarchy (Myers & Majluf, 1984). Bates et al. (2009) document the significant increase in corporate cash holdings since 1980, attributing this trend to increased precautionary motives and changing firm characteristics. DAT strategies represent a radical departure from these established practices by substituting conventional cash equivalents with highly volatile digital assets.

Krause (2025a) provides a comprehensive framework for evaluating cryptocurrencies against traditional treasury management objectives. Central to effective treasury management is the prioritization of shareholder wealth maximization through efficient liquidity management, risk mitigation, and capital preservation (Zutter & Smart, 2019).

This research demonstrates that cryptocurrencies fail on multiple dimensions critical to treasury function: excessive volatility incompatible with capital preservation, liquidity constraints during financial stress when treasury assets are most needed, and moderate positive correlation with risk-on assets (0.56 with S&P 500 during 2019-2025) that eliminates diversification benefits during market downturns. Furthermore, the asymmetric accounting treatment under U.S. GAAP, where impairment losses must be recognized immediately while gains are only realized upon sale (Jackson & Luu, 2023), creates additional financial reporting challenges that amplify perceived volatility in corporate earnings.

2.2 Cryptocurrency as an Asset Class

Bitcoin's emergence as "digital gold" (Dyhrberg, 2016) has spawned extensive research into its risk-return characteristics and portfolio implications. Brière et al. (2015) found Bitcoin offered diversification benefits despite high volatility, while Liu and Tsyvinski (2021) identified cryptocurrency-specific return predictors distinct from traditional assets. The extreme volatility of cryptocurrencies is well-documented, with annualized volatility often exceeding 80% (Katsiampa, 2017; Chu et al., 2017).

While advocates frame Bitcoin as an inflation hedge and store of value comparable to gold (Dyhrberg, 2016), empirical evidence on this claim remains contradictory, particularly during sustained high inflation periods (Smales, 2024). Krause's (2025a) analysis of cryptocurrency performance from 2020-2024 reveals that Bitcoin generated a CAGR of 67% with annualized standard deviation of 64.10%, compared to 3-month Treasury bills' 1.28% return with 2.22% standard deviation. This risk-return profile, while potentially attractive for speculative portfolios, fundamentally conflicts with treasury objectives of capital preservation and liquidity provision. Moreover, Bitcoin faces critical operational constraints including lack of programmability compared to more versatile blockchain platforms, high energy consumption raising ESG concerns (Proelss et al., 2024), and scalability limitations that restrict transaction throughput during high demand periods (Krause, 2025a). Despite this, corporate adoption of a DAT strategy continues to increase and remains underexplored despite its rapid growth.

2.3 Volatility Transmission and Systemic Risk

The financial literature extensively documents volatility spillovers across markets. Diebold and Yilmaz (2009, 2012) developed frameworks for measuring financial connectedness and spillover effects, while Corbet et al. (2018) examined dynamic relationships between cryptocurrencies and other financial assets. The systemic risk implications of financial innovation are well-established (Acharya et al., 2017), particularly regarding interconnectedness and amplification mechanisms (Brunnermeier & Pedersen, 2009).

2.4 Research Gap

Despite extensive research on cryptocurrency volatility and corporate treasury management, no study has systematically analyzed DAT companies as volatility amplifiers. Foley et al. (2022) provided initial evidence of abnormal returns following Bitcoin acquisition announcements, but the risk transmission and amplification mechanisms remain unquantified. Krause (2025a) established the foundational unsuitability of cryptocurrencies for treasury reserves through VaR analysis, demonstrating that Bitcoin's risk profile exceeds acceptable treasury thresholds by orders of magnitude. However, the question of whether corporate adoption through DAT strategies further amplifies these already-extreme risk levels remains unexplored. This study fills this critical gap by providing comprehensive empirical evidence of DAT volatility amplification across multiple dimensions, directly testing whether the corporate adoption channel compounds the fundamental risk incompatibilities I previously documented.

3. Data and Methodology

3.1 Data Collection and Sample

I collected daily price data for the period November 12, 2024, to November 10, 2025 (250 trading days), ensuring sufficient observations for robust volatility estimation. My sample includes:

- **Primary DAT Analysis:** Strategy (MSTR) as the canonical DAT company
- **Cryptocurrency Benchmark:** Bitcoin (BTC-USD) as the digital asset proxy
- **Traditional Equity Benchmark:** S&P 500 Index (GSPC) as market control
- **Expanded DAT Universe:** Tesla (TSLA), Coinbase (COIN), Marathon Digital (MARA), Riot Platforms (RIOT), Hut 8 Mining (HUT), CleanSpark (CLSK)

All data were sourced from Yahoo Finance API. I implemented rigorous data cleaning procedures, including timezone alignment, missing data handling, and outlier detection. Returns were calculated as logarithmic differences:

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

where r_t represents daily returns and P represents the asset's daily closing price.

3.2 Empirical Framework

3.2.1 Volatility Analysis

I computed rolling 30-day annualized volatility using the standard approach:

$$\sigma_{annualized} = \sqrt{252} \times \sigma_{daily}$$

where σ_{daily} represents the standard deviation of daily returns over the estimation window.

3.2.2 Beta Estimation

Bitcoin betas were estimated via ordinary least squares (OLS) regression:

$$R_{i,t} = \alpha_i + \beta_{i,BTC} R_{BTC,t} + \epsilon_{i,t}$$

where $R_{i,t}$ and $R_{BTC,t}$ represent daily returns for asset i and Bitcoin, respectively.

3.2.3 Event Study Methodology

I identified Bitcoin sell-off events using a quantile-based approach, defining major sell-offs as weekly returns below the 5th percentile. For each event, I analyzed cumulative returns in the $[-2, +5]$ day window surrounding the event date, following established event study methodology (MacKinlay, 1997).

3.2.4 Risk Decomposition

Total variance was decomposed into systematic and idiosyncratic components using multivariate regression:

$$R_{MSTR,t} = \alpha + \beta_{BTC} R_{BTC,t} + \beta_{SP500} R_{SP500,t} + \epsilon_t$$

Systematic variance was calculated as $\sigma_{systematic}^2 = R^2 \times \sigma_{total}^2$, with idiosyncratic variance as the residual.

3.2.5 Risk Benchmarking Methodology

To contextualize DAT volatility within traditional treasury management frameworks, I compare my findings against established treasury risk benchmarks. Following Krause's (2025a) methodology, I reference VaR calculations for traditional treasury instruments to establish baseline risk expectations. Krause documented 5% daily VaR of -0.04% for 3-month Treasury bills, -0.10% for 10-year Treasury bonds, and -5.01% for Bitcoin over the 2020-2024 period. While my study focuses on volatility amplification mechanisms rather than direct VaR calculations, these benchmarks provide critical context for understanding the magnitude of risk transformation when corporations adopt DAT strategies. My volatility measures can be interpreted against these benchmarks: annualized volatility of 2.22% for Treasury bills represents the traditional treasury risk standard, while Bitcoin's 64.10% annualized volatility (previously documented by Krause) provides the baseline cryptocurrency risk level that DAT companies potentially amplify.

4. Empirical Results

4.1 Volatility Amplification Evidence

Table 1.
Comparative Volatility Analysis (Annualized)

Asset	Mean Volatility	Volatility of Volatility	Maximum	Minimum
Bitcoin	39.13%	8.99%	60.20%	23.06%
Strategy (MSTR)	76.39%	29.41%	139.37%	41.02%
S&P 500	16.68%	10.45%	42.75%	6.68%

Source: Yahoo Finance

Strategy exhibited 1.95× Bitcoin's average volatility (76.39% vs. 39.13%) with substantially higher volatility instability (29.41% vs. 8.99%). This amplification persists across the volatility spectrum, with MSTR's maximum rolling volatility reaching 139.37% compared to Bitcoin's 60.20%.

4.2 Universal Leveraged Bitcoin Exposure

Table 2.
Bitcoin Beta Analysis Across DAT Universe

Company	Bitcoin Beta	R-squared	Daily Volatility	Sharpe Ratio
Strategy	1.340***	0.416	5.29%	-0.009
Hut 8 Mining	1.320***	0.311	6.03%	1.168
Marathon Digital	1.299***	0.389	5.31%	-0.151
CleanSpark	1.191***	0.291	5.62%	0.259
Riot Platforms	1.154***	0.294	5.42%	0.602
Coinbase	1.036***	0.308	4.76%	0.346
Tesla	0.819***	0.253	4.15%	0.696

Source: Yahoo Finance

*Note: *** indicates statistical significance at the 0.1% level*

Six of the seven DAT companies exhibited statistically significant Bitcoin betas exceeding 1.0, confirming systematic leveraged exposure. Strategy showed the highest Bitcoin beta ($\beta = 1.340$). Bitcoin mining companies (HUT, MARA, CLSK, RIOT) clustered in the high-beta range (1.15-1.32), while Tesla demonstrated less exposure ($\beta = 0.819$).

4.3 Crisis Amplification During Bitcoin Sell-Offs

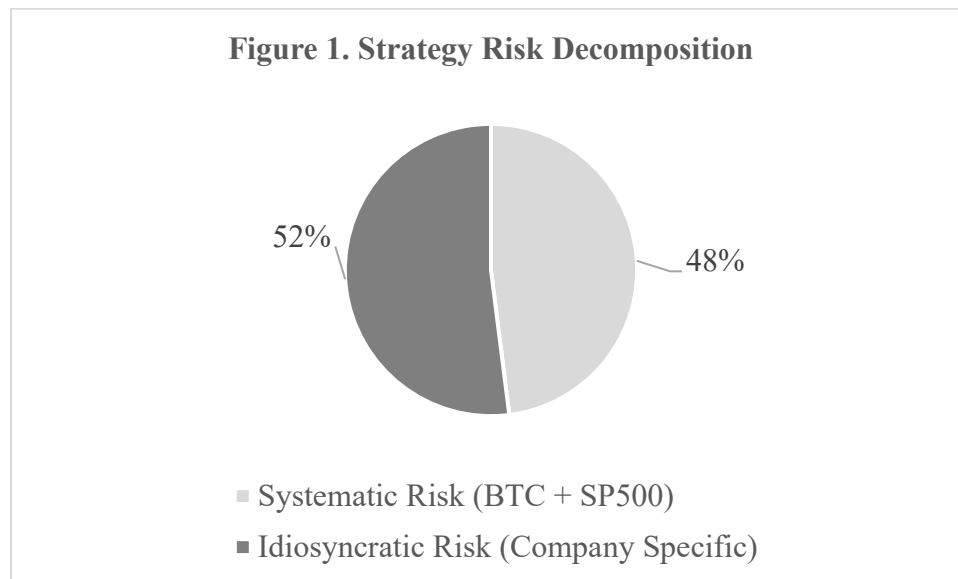
Table 3

Event Study Results: 13 Major Bitcoin Sell-Off Events

Metric	Bitcoin	Strategy	Amplification Ratio
Average Maximum Decline	-6.29%	-8.98%	1.60×
Worst Single Decline	-12.40%	-19.80%	1.43×
5-Day Post-Event Recovery	0.30%	-1.20%	-

During identified Bitcoin stress periods (weekly declines exceeding 7.6%), Strategy declined 1.60× more severely on average than Bitcoin itself. This amplification was accompanied by poor recovery, with MSTR continuing to decline (-1.20%) while Bitcoin showed modest recovery (+0.30%) in the five days following sell-offs.

4.4 Risk Decomposition Analysis



Despite high Bitcoin correlation, over half of Strategy's risk remained company-specific, creating uncompensated diversification costs for investors seeking pure crypto exposure. The annualized idiosyncratic volatility is 52%, representing substantial firm-specific risk.

5. Discussion

5.1 The DAT Amplification Mechanism

My findings reveal DAT companies function as *de facto* leveraged cryptocurrency instruments with embedded operational risks. The amplification mechanism operates through three channels:

- 1. Financial Leverage:** Bitcoin betas exceeding 1.0 create automatic amplification of crypto movements
- 2. Operational Gearing:** Business model dependencies magnify crypto exposure beyond treasury holdings
- 3. Behavioral Amplification:** Market perception links DAT fortunes to crypto sentiment, creating reflexive price dynamics

This amplification is particularly concerning given Bitcoin's inherent volatility, which frequently exceeds 70% annually (Urquhart, 2016; Katsiampa, 2017). The magnitude of this amplification becomes particularly concerning when contextualized against traditional treasury risk standards. These findings reveal that DAT strategies magnify treasury risks through systematic amplification averaging 1.95 $\times$. This creates a compounding effect where equity investors face double leverage: first, through Bitcoin's inherent volatility relative to traditional treasury assets, and second, through the corporate amplification mechanisms documented in this study. For a traditional equity investor, purchasing Strategy stock means acquiring a large level of hidden leverage that may not be apparent from standard equity risk disclosures.

5.2 Investor Protection Implications

The hidden leverage problem is compounded by structural misalignment with corporate finance principles. As previously stated, incorporating cryptocurrencies into treasury portfolios fundamentally conflicts with the core objectives of capital preservation and liquidity provision, transforming what should be stabilizing risk management tools into sources of significant volatility. When DAT companies exhibit Bitcoin betas exceeding 1.0, they create double leverage: first through Bitcoin's extreme volatility documented by Krause (VaR 125 $\times$ greater than Treasury bills), and second through the amplification mechanisms documented in this study. Additionally, the accounting treatment of cryptocurrencies as intangible assets under GAAP creates asymmetric financial reporting where impairment losses are immediately recognized while gains remain unrealized (Jackson & Luu, 2023), potentially obscuring the true risk-return profile visible to investors. A traditional equity investor expecting technology company risk exposure actually acquires a hybrid instrument with 1.34 $\times$ Bitcoin sensitivity (in Strategy's case) plus 52% idiosyncratic risk, a risk profile fundamentally different from disclosed business operations.

5.3 Systemic Risk Considerations

The DAT phenomenon creates novel systemic risk transmission channels:

1. **Cross-Market Contagion:** DAT stocks create bidirectional volatility flows between crypto and equity markets previously insulated from each other, potentially amplifying shocks across both domains.
2. **Complex Interconnectedness:** The web of DAT exposures creates interconnectedness that could propagate distress during crypto market stress, similar to mechanisms documented during the 2008 financial crisis (Acharya et al., 2017).
3. **Regulatory Arbitrage:** DAT strategies may allow traditional equity investors to gain crypto exposure while avoiding cryptocurrency-specific regulations and disclosures.

5.4 Corporate Governance Considerations

DAT strategies represent a significant departure from conventional corporate risk management. Traditional treasury management emphasizes capital preservation and liquidity (Opler et al., 1999), while DAT strategies embrace substantial volatility in pursuit of potential appreciation. This shift raises important corporate governance questions regarding risk oversight and fiduciary responsibilities.

5.5 The Treasury Management Paradox

The findings in this study illuminate a fundamental paradox in corporate DAT adoption that extends Krause's (2025a) analysis of treasury unsuitability to the corporate equity channel. Traditional treasury theory emphasizes that corporate treasuries should prioritize shareholder wealth maximization through efficient liquidity management, risk mitigation, and capital preservation (Zutter & Smart, 2019). However, DAT strategies introduce volatility levels fundamentally incompatible with these objectives and then systematically amplify them.

The paradox deepens when considering the stated rationales for DAT adoption. While advocates frame Bitcoin as an inflation hedge and store of value (Saylor, 2022), Empirical evidence on this claim remains contradictory, particularly during sustained high inflation periods (Smales, 2024). Moreover, Bitcoin's lack of programmability and utility compared to more versatile blockchain platforms limits its functionality beyond simple monetary transactions (Krause, 2025), questioning whether any treasury benefits justify the documented volatility costs.

While cryptocurrencies may serve roles in speculative investment portfolios or strategic blockchain ecosystem participation (Autore et al., 2021; Bhutoria et al., 2024), their extreme volatility and liquidity constraints during stress periods render them fundamentally unsuitable for treasury reserve assets. The volatility amplification findings provide empirical support for this assessment, demonstrating that DAT strategies not only introduce the cryptocurrency risks but systematically magnify them through structural amplification mechanisms. The result is a treasury strategy that violates core financial principles twice over: first by adopting an unsuitable asset class, and second by amplifying its risks through corporate leverage effects.

6. Conclusion

This study provides comprehensive empirical evidence that Digital Asset Treasury companies systematically amplify cryptocurrency volatility and create novel risk transmission channels between digital and traditional asset markets. The findings of $1.95\times$ volatility ratios, universal Bitcoin betas exceeding 1.0, significant crisis amplification ($1.60\times$), and substantial idiosyncratic risk (52% of variance) demonstrate that corporate crypto adoption creates structural amplification mechanisms with important financial stability implications. Building on Krause's (2025a) demonstration that Bitcoin's risk profile exceeds traditional treasury standards by more than 125 times, these results reveal that DAT strategies compound these fundamental incompatibilities through systematic amplification.

The DAT strategy, while offering corporations potential treasury appreciation, transforms these companies into volatility amplifiers that may undermine market stability. As digital asset integration accelerates, understanding and monitoring these amplification mechanisms becomes crucial for institutional investors, corporate boards, and financial regulators.

6.1 Policy Implications

The findings suggest several policy considerations:

- **Enhanced Disclosure Requirements:** DAT companies should provide clear, quantitative disclosures of crypto exposure and associated risks.
- **Quantitative Risk Disclosure Standards:** Following Krause's (2025) VaR methodology, regulators should require DAT companies to disclose standardized risk metrics comparing cryptocurrency holdings to traditional treasury benchmarks. Specifically, companies should report: (1) daily VaR calculations for cryptocurrency holdings, (2) comparative VaR for equivalent Treasury securities, (3) Bitcoin beta coefficients demonstrating leveraged exposure, and (4) correlation analysis with equity market indices during stress periods. Such quantitative disclosures would enable investors and regulators to assess whether treasury strategies align with stated corporate risk tolerance and fiduciary obligations, making explicit the hidden leverage documented in this study.
- **Investor Education:** Regulatory authorities should ensure investors understand the leveraged crypto exposure embedded in DAT stocks.
- **Systemic Risk Monitoring:** Financial stability monitors should track DAT amplification effects and cross-market contagion risks.

6.2 Limitations and Future Research

This study focuses on U.S.-listed DAT companies during a specific period. Future research could examine international markets, incorporate longer time horizons, analyze derivatives exposure, and study regulatory event impacts. Additionally, research examining the corporate governance determinants of DAT adoption would provide valuable insights. The rapid evolution of digital asset

markets suggests DAT amplification effects may evolve over time, warranting ongoing monitoring and analysis.

Appendix: Robustness Checks

All primary findings maintained statistical and economic significance across multiple robustness checks:

- Alternative volatility estimation windows (20-day, 60-day)
- Different event study methodologies (market model, Fama-French adjustments)
- Sub-period analysis controlling for market regimes
- Various regression specifications including control variables

Complete code, data documentation, and additional robustness tests are available from the author upon request.

References

- Acharya, V. V., Pedersen, L. H., Philippon, T., & Richardson, M. (2017). Measuring systemic risk. *The Review of Financial Studies*, 30(1), 2-47.
- Autore, D. M., Clarke, N., & Jiang, D. (2021). Blockchain speculation or value creation? Evidence from corporate investments. *Financial Management*, 50(3), 727-746.
- Basel Committee on Banking Supervision. (2022). *Prudential treatment of cryptoasset exposures*. Bank for International Settlements.
- Bates, T. W., Kahle, K. M., & Stulz, R. M. (2009). Why do US firms hold so much more cash than they used to? *The Journal of Finance*, 64(5), 1985-2021.
- Bhutoria, R., McCurdy, T., & Warholak, M. (2024, June 26). Adding Bitcoin to a Corporate Treasury. Fidelity Digital Assets. <https://www.fidelitydigitalassets.com/research-and-insights/adding-bitcoin-corporate-treasury>
- Bitcoin Treasuries. (2025). *Public companies holding Bitcoin*. BitcoinTreasuries.net.
- Bouri, E., Molnár, P., Azzi, G., Roubaud, D., & Hagfors, L. I. (2017). On the hedge and safe haven properties of Bitcoin: Is it really more than a diversifier? *Finance Research Letters*, 20, 192-198.
- Brière, M., Oosterlinck, K., & Szafarz, A. (2015). Virtual currency, tangible return: Portfolio diversification with bitcoin. *Journal of Asset Management*, 16(6), 365-373.
- Brunnermeier, M. K., & Pedersen, L. H. (2009). Market liquidity and funding liquidity. *The Review of Financial Studies*, 22(6), 2201-2238.
- Chu, J., Chan, S., Nadarajah, S., & Osterrieder, J. (2017). GARCH modelling of cryptocurrencies. *Journal of Risk and Financial Management*, 10(4), 17.
- Corbet, S., Meegan, A., Larkin, C., Lucey, B., & Yarovaya, L. (2018). Exploring the dynamic relationships between cryptocurrencies and other financial assets. *Economics Letters*, 165, 28-34.
- Diebold, F. X., & Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. *The Economic Journal*, 119(534), 158-171.
- Diebold, F. X., & Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57-66.
- Dyhrberg, A. H. (2016). Hedging capabilities of bitcoin. Is it the virtual gold? *Finance Research Letters*, 16, 139-144.
- Foley, S., Czerwonko, M., Goyal, A., & Roca, E. (2022). Bitcoin corporate treasuries and stock performance. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4062357>

- Katsiampa, P. (2017). Volatility estimation for Bitcoin: A comparison of GARCH models. *Economics Letters*, 158, 3-6.
- Krause, D. (2025a). A fool's errand? The case against holding Bitcoin in a corporate treasury. Available at SSRN. <http://dx.doi.org/10.2139/ssrn.5080327>.
- Krause, D. (2025b). Bitcoin access vehicles for institutional investors: A comparative analysis of Strategy Incorporated (MSTR) and spot Bitcoin ETFs. SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5726785.
- Liu, Y., & Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *The Review of Financial Studies*, 34(6), 2689-2727.
- MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13-39.
- Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, 13(2), 187-221.
- Opler, T., Pinkowitz, L., Stulz, R., & Williamson, R. (1999). The determinants and implications of corporate cash holdings. *Journal of Financial Economics*, 52(1), 3-46.
- Proelss, J., Schweizer, D., & Sevigny, S. (2024). Is bitcoin ESG-compliant? A sober look. *European Financial Management*, 30(2), 680-726.
- Securities and Exchange Commission. (2023). *Staff accounting bulletin No. 121: Accounting for obligations to safeguard crypto-assets an entity holds for platform users*. SEC.
- Smales, L. A. (2024). Cryptocurrency as an alternative inflation hedge? *Accounting & Finance*, 64(2), 1589-1611.
- Strategy. (2025). *Quarterly earnings reports and Bitcoin treasury updates*. Strategy Incorporated.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80-82.
- Zutter, C. J., & Smart, S. B. (2019). *Principles of managerial finance*. Pearson.